Week 01_Module 02_Part 01

📊 Module 02: Probability Basics for Machine Learning

Topics Covered

• Events, sample space, and basic probability rules
• Conditional probability and independence
• Bayes' theorem (intuitive and formula-level) and base-rate effects
• Sensitivity, specificity, false positive/negative rates, PPV, and NPV
• Class imbalance and its impact on interpreting errors

What Is Probability and Why It Matters for ML

Probability is the mathematics of uncertainty — it tells us how likely an event is to occur.
Machine Learning relies on probability because models don't make absolute decisions; they estimate the likelihood of outcomes based on data.

Examples:

- Probability that an email is spam = 0.9 → model is 90 % confident it's spam.
- Probability that a loan will default = 0.2 → 20 % risk of default.
- Probability that a patient has diabetes = 0.6 → moderate chance requiring further tests.

Probability lets ML systems **handle noise, uncertainty, and incomplete information** while still making reasoned predictions.

1️⃣ Events, Sample Space, and Basic Rules

- **Event:** A specific outcome (e.g., rolling an even number on a die).
- **Sample Space:** All possible outcomes (e.g., {1, 2, 3, 4, 5, 6}).
- **Rule:** P(Event) = Favorable outcomes / Total outcomes.

Example: P(Even) = 3 / 6 = 0.5.

In ML, an event could be "customer buys product" or "model predicts positive."

2️⃣ Conditional Probability and Independence

Conditional probability is the probability of A *given that* B has occurred.

Example: P(Person buys ice cream | Hot day).
It answers how one factor changes the likelihood of another.

In ML, conditional probability helps find relationships like
P(Class | Features) — the chance of a label given input data.

Independence: Events A and B are independent if P(A | B) = P(A).

Naïve Bayes classifier assumes this independence for simplicity, even though real data rarely follows it perfectly.

3️⃣ Bayes' Theorem and Base-Rate Effects

Bayes' Theorem lets us **update our beliefs when new evidence appears.**
Formula: P(A | B) = [P(B | A) × P(A)] / P(B)

Intuition:

- Start with a belief (prior).
- Observe new data (likelihood).
- Update belief (posterior).

Example: A disease test is 99 % accurate, but if the disease affects only 1 in 10 000 people, most positive results are still false alarms. That's the **base-rate effect** — even accurate tests can mislead when the event is rare.

In ML, this teaches us to adjust our expectations based on how common each class is.

4️⃣ Sensitivity, Specificity, and Error Rates

These concepts come from the **confusion matrix**, a key tool for evaluating models.

- **Sensitivity (True Positive Rate):** Correctly detecting positives.
- **Specificity (True Negative Rate):** Correctly detecting negatives.
- **False Positive:** Model predicts positive when it's actually negative.
- **False Negative:** Model predicts negative when it's actually positive.
- **PPV (Positive Predictive Value):** How often a positive prediction is correct.
- **NPV (Negative Predictive Value):** How often a negative prediction is correct.

These metrics help decide what matters most in a task — catching every positive case (sensitivity) or avoiding false alarms (specificity).

5️⃣ Class Imbalance and Its Impact

Many real-world datasets are **imbalanced** — one class dominates.

Example: In 10 000 transactions, only 50 are fraudulent (0.5 %).
A model that always predicts "no fraud" is 99.5 % accurate — and completely useless.
That's why accuracy alone isn't enough.
We use probabilistic metrics like PPV, NPV, and Bayesian reasoning to judge how well a model really handles minority classes.

Understanding these concepts prepares you for later modules on **model evaluation, precision, recall, and F1-score.**

🎯 Module Goal

By the end of Module 02, you should be able to:
✅ Explain probability concepts in ML context.
✅ Apply Bayes' Theorem intuitively.
✅ Interpret confusion matrix metrics (sensitivity, specificity, PPV, NPV).
✅ Recognize how class imbalance distorts model accuracy.

Remember: Machine Learning is not about certainty — it's about **making smart decisions under uncertainty.**